Unit 1: Biological Bases of Behavior Part II

Myers: 16, 18, 20, 21, 22, 23, 24
AP Weighting: 15-25%

In this half of Unit 1, you will learn about sleep, dreams, altered states of consciousness, and each of the different sensation systems within our body. This unit provides a comprehensive understanding of how biology and psychology intersect to shape human experiences and actions.

Topic 1.5: Sleep

Learning Target 1.5.A: Explain how the sleep/wake cycle affects behavior and mental processes throughout the day and night.

I. Consciousness:

A. Our consciousness has 3 main functions:

1. Monitoring the environment (awareness of what's around us).
2. Controlling behavior (planning, focusing, directing actions).
3. Sorting & making sense of info (selective attention, memory, reasoning).

II. Biological Rhythms:

A. Circadian Rhythm: Biological cycle of about 24 hours.

1. What is one thing our circadian rhythm regulates? Sleep/wake cycle (about 24 hrs); also body temp & hormones.
2. Jet Lag: Circadian rhythm out of sync after fast travel across time zones; causes tiredness, trouble sleeping.

3. Shift Work Sleep Disorder: Work schedule clashes with the body clock, causing insomnia and daytime sleepiness.

B. Ultradian Rhythm: Cycle shorter than 24 hrs (e.g., 90-minute sleep cycles, hunger).

C. Infradian Rhythm: Cycle longer than 24 hrs (e.g., menstrual cycle, seasonal changes).

III. Sleep

A. What are the key functions of sleep, and why is it important? Restores/repairs the body, consolidates memory, supports growth, mood, immunity and learning.

B. What part of your brain regulates sleep? How? Hypothalamus (suprachiasmatic nucleus) – uses light cues from the eyes to set the body clock.

1. Which endocrine gland is associated with sleep? Pineal gland.

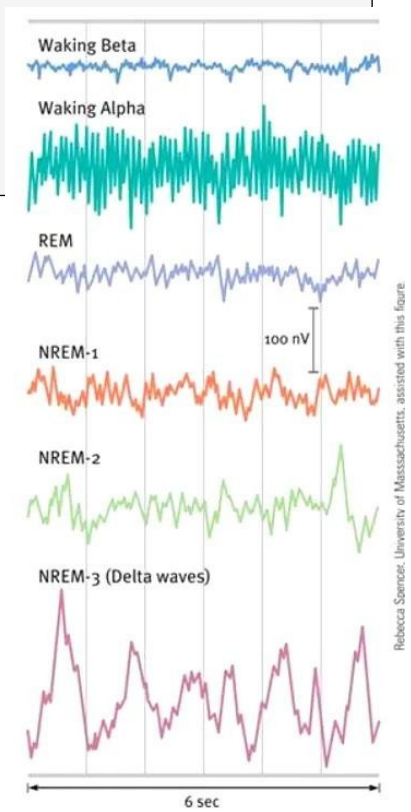
2. Which hormone does that endocrine gland secrete? Melatonin.

C. Sleep Stages

1. What is the key difference between NREM stages and the REM stage? NREM = quiet, no vivid dreams, muscles can move; REM = rapid eye movement, vivid dreams, muscles paralyzed, brain very active.

2. How long does it take for us to cycle through the four sleep stages? About 90 minutes.

Stage	Wave Name	Wave Characteristics	Common Behaviors of This Stage
Awake	Beta	Small, fast, irregular	Alert, active, thinking, focusing
Awake, but Relaxed	Alpha	Slower, larger, regular	Relaxed, calm, drowsy, eyes closed
NREM-1	Theta	Slower, irregular; bursts called hypnagogic	Light sleep, easily awakened Hypnagogic Sensations: Vivid sensory images, feeling of falling or floating, sudden muscle jerks.
NREM-2	Theta + sleep spindles & K-complexes	Slow with brief bursts of rapid activity	Breathing/heart rate slow; harder to wake; memory consolidation
NREM-3	Delta	Large, slow, high amplitude	Deep sleep; hardest to wake; growth hormone, night terrors, sleepwalking
REM	Beta-like (fast, sawtooth)	Fast, irregular, like awake	Vivid dreams, rapid eye movements, muscle paralysis Why is REM sleep often called "paradoxical sleep"? Brain is highly active (like when awake) but the body is paralyzed and still.



D. What are some effects of sleep loss? Poor memory/attention, irritability, weaker immunity, weight gain, slower reactions, higher risk of accidents.

1. REM Rebound: Extra REM sleep after REM deprivation.

E. Sleep Disorders

1. Insomnia: Trouble falling or staying asleep.

2. Narcolepsy: Sudden uncontrollable sleep attacks, often straight into REM.

3. Sleep Apnea: Breathing repeatedly stops during sleep, causing awakenings and gasping.

4. Night Terrors: Sudden terror, screaming, and arousal with no memory of it later.

   a) In which sleep stage do they occur? NREM-3 (stage 3).

5. REM Sleep Behavior Disorder: Muscle paralysis fails, so person acts out dreams.

6. Somnambulism: Sleepwalking (NREM-3).

IV. Dreaming

A. In which sleep stage do dreams occur? Mostly REM (also some in NREM).

B. Dream Theories:

1. Freud's Wish Fulfillment: Dreams are disguised expressions of unconscious wishes/desires.

a) Manifest Content: The actual storyline of the dream that you remember.

b) Latent Content: The hidden, symbolic meaning of the dream.

2. Consolidation Theory Dreams help process and store the day's memories and learning.

3. Activation-Synthesis: Dreams are the brain's attempt to make sense of random neural activity during REM.

Topic 1.6: Sensation

Learning Target 1.6.A: Explain how the process of sensation is related to behavior and mental processes.

I. **What is sensation? How does it differ from perception?** Sensation = receptors detect stimuli and send info to the brain; perception = brain organizes and interprets that info.

C. **Bottom-Up Processing:** Starts with sensory input (details) and builds up to perception.

D. **Top-Down Processing:** Uses experience and expectations to interpret sensory input.

E. **Synesthesia:** Senses blend, e.g., seeing colors when hearing sounds or reading numbers.

V. **Transduction:**

A. **Step 1: Reception:**

1. External Stimuli: Physical energy from the environment (light, sound, chemicals, pressure).

2. Sensory Receptors: Specialized cells that detect stimuli and respond to them.

B. **Step 2: Transformation:** Receptors convert stimulus energy into neural impulses (electrical signals).

C. **Step 3: Transmission:** Neural impulses travel to the brain via the nervous system.

VI. Thresholds

A. **Absolute Threshold:** Minimum stimulation needed to detect a stimulus 50% of the time.

B. **Difference Threshold:** Minimum difference between two stimuli that is noticed 50% of the time.

1. Just Noticeable Difference (JND): The smallest change in a stimulus that can be detected.

2. Weber's Law: To be noticed as different, two stimuli must differ by a constant proportion, not amount.

3. How does Weber's Law help explain JND? The bigger/stronger the original stimulus, the larger the change needed to notice it (e.g., a loud sound needs a bigger increase).

C. **Signal Detection Theory:** Predicts when we detect a faint stimulus among background noise.

1. Depends on two conditions: Strength of the signal vs. noise, and the person's experience, expectations, motivation and alertness.

D. **Subliminal Stimuli:** Stimuli below the absolute threshold that can't be consciously detected.

1. Backmasking: Hidden messages in music played backward; claimed to be subliminal (little evidence it works).

E. Sensory Adaptation: Decreased sensitivity to a constant, unchanging stimulus.

1. Why is this evolutionarily advantageous? Frees attention to notice important changes (like danger) instead of constant stimuli.

F. Sensory Habituation: Brain learns to ignore a repeated stimulus (a decrease in response over time).

G. Sensory Interaction: One sense influences another (e.g., smell affects taste).

Learning Target 1.6.B: Explain how the structures and functions of the *visual* sensory system relate to behavior and mental processes.

I. Understand these anatomical structures in the eye:

Structure	Function
Cornea	Clear outer covering; protects the eye and bends (refracts) light into it.
Pupil	Small opening in the center of the iris; lets light in.
Iris	Muscle ring that controls pupil size (more/less light) and gives eye color.
Lens	**Accommodation -** Lens changes shape to focus on near or far objects. **Dysfunctions:** Myopia, hyperopia, presbyopia (stiff lens with age), cataracts (cloudy lens).

Structure	Function
Retina	**Location:** Back inner surface of the eye; contains rods, cones and other cells; focused image lands here.
Optic Nerve	**Blind Spot:** Where the optic nerve leaves the eye; no receptors, so no vision there. **Optic Chiasm:** Where optic nerves cross; left visual field goes to right brain and vice versa.

VII. What happens in the retina?

A. First Layer

1. Rods: Detect black, white, gray and dim light; peripheral vision; located around the edges of retina; no color.

    a) **Dark Adaptation:** Retina's increased sensitivity in dim light (takes ~20-30 min).

2. Cones:. Detect color and fine detail in bright light; concentrated in the fovea.

B. Second Layer

1. Bipolar Cells: Receive signals from rods/cones and pass them to ganglion cells.

C. Third Layer

1. Ganglion Cells: Their axons form the optic nerve and carry the signal out of the eye.

VIII. What happens to the neural impulse after it exits the eye?

A. Thalamus Relay station; sends visual info to the visual cortex.

B. Where in the brain are visual impulses processed? Visual cortex in the occipital lobe.

1. Feature Detectors:. Neurons that respond to specific features (edges, lines, angles, movement).

2. Supercell Clusters:. Groups of neurons that respond to complex combos of features (e.g., faces).

IX. How Do We Process Visual Information?

Part of the Wave	Define	Visual Element
Wavelength	Distance from one wave peak to the next.	Hue (color)
Amplitude	Height of the wave (its intensity).	Brightness

X. Color Vision Theories

A. **Young-Helmholtz Trichromatic Color Theory** Retina has 3 cone types (red, green, blue); mixing their activity produces all colors.

B. **Opponent Process Theory** Color is processed in opposing pairs: red-green, blue-yellow, black-white.

1. Afterimage: Image seen after staring then looking away (e.g., stare at green, see red) as the fatigued color cells rest.

C. **How do we reconcile these two theories?** Both are right: trichromatic works at the retina (cones); opponent-process works after, in the ganglion cells/thalamus.

XI. Common Visual Disorders

A. **Nearsightedness (Myopia):** Can see near but not far; image focuses in front of the retina.

B. Farsightedness (Hyperopia): Can see far but not near; image focuses behind the retina.

C. Astigmatism: Irregularly shaped cornea/lens causes blurry or distorted vision.

D. What is the difference between dichromatism and monochromatism?. Dichromatism = only 2 cone types work (e.g., red-green colorblindness); monochromatism = only 1 (or none) works, so no color.

E. How does brain damage result in disorders like *prosopagnosia* and *blindsight*?
Prosopagnosia: damage to the temporal lobe (face area) so faces can't be recognized.
Blindsight: damage to the visual cortex (occipital lobe) so a person says they can't see but still responds to objects.

Learning Target 1.6.C: Explain how the structures and functions of the *auditory* sensory system relate to behavior and mental processes.

I. Sound Waves

F. How do air pressure waves become sound? Vibrating objects create pressure waves in air that the ear detects and the brain interprets as sound.

G. Frequency Number of wavelengths per second (Hertz).

1. Pitch: How high or low a sound is; determined by frequency.

a) What does a high frequency sound like? High pitch (e.g., whistle).

b) What does a low frequency sound like? Low pitch (e.g., bass).

H. Amplitude: Height of the wave; determines loudness.

1. What does a large amplitude sound like? Loud.

2. What does a small amplitude sound like? Soft/quiet.

XII. How does the ear transform sound into neural messages?

A. Outer Ear (Pinna): Visible part; collects and funnels sound waves into the ear.

1. Auditory Canal Channels the sound waves to the eardrum.

2. Ear Drum (Tympanic Membrane) Membrane that vibrates when hit by sound waves.

B. Middle Ear: Amplifies the vibrations.

3. Ossicles (Name all 3) Hammer (malleus), anvil (incus), stirrup (stapes).

C. Inner Ear:. Converts vibrations into neural signals.

4. Oval Window: Membrane the stirrup vibrates against, sending vibrations into the cochlea.

5. Cochlea: Fluid-filled, snail-shaped tube containing the hearing receptors.

6. How does the sound wave move through the inner ear? Stirrup → oval window → cochlear fluid ripples → basilar membrane moves.

7. How does transduction occur in the inner ear? Fluid ripples bend hair cells, which trigger neural impulses.

   a) *Basilar Membrane:* Membrane in the cochlea that holds the hair cells and vibrates with the fluid.

b) *Hair Cells:* Auditory receptors; bending creates neural signals sent to the auditory nerve.

D. How does the brain detect loudness? By the number of hair cells activated: louder sounds activate more cells (and they fire faster).

E. How do place theory and frequency theory work together to explain how we hear pitch? Place theory: pitch depends on where on the basilar membrane it vibrates (high pitches). Frequency theory: neurons fire at the same rate as the sound (low pitches). Together they cover the full range of pitch.

F. Volley Principle Groups of neurons take turns firing, so together they can match higher frequencies.

G. How do we locate sounds? Draw the concept of sound localization. Brain compares the sound's arrival time and loudness at the two ears. (Sketch: head with sound source off to the right → reaches right ear first and louder, left ear slightly later and softer.)

XIII. How does the nerve impulse move out of the ear?

A. Auditory Nerve: Carries neural impulses from the cochlea to the brain.

B. Thalamus: Relays auditory info to the auditory cortex.

C. Where in the brain are auditory impulses processed? Auditory cortex in the temporal lobe.

XIV. Hearing Loss

A. Sensorineural Nerve deafness: damage to hair cells or the auditory nerve (most common, permanent).

1. What is the problem with headphones? Loud volume damages hair cells, which don't regrow.

B. Conduction Damage to the eardrum or ossicles, so vibrations don't reach the cochlea.

1. How does a cochlear implant work? It converts sounds into electrical signals and stimulates the auditory nerve directly through electrodes in the cochlea.

Learning Target 1.6.D: Explain how the structures and functions of the *chemical* sensory system relate to behavior and mental processes.

I. Gustation: Sense of taste.

C. What are the five basic tastes we can detect? Indicate at least one taste substance for each taste (ex. Sucrose for sweet) Sweet (sucrose), sour (acids/citric acid), salty (sodium chloride), bitter (quinine/caffeine), umami (glutamate/MSG).

D. What are the survival functions of each basic taste? Sweet = energy source; sour = spoiled/unripe food; salty = sodium balance; bitter = toxins/poisons; umami = proteins.

E. How do we actually taste food? Chemicals in food dissolve in saliva and trigger taste receptors on the tongue.

1. Fungiform Papillae: Bumps on the tongue that hold the taste buds.

a) *Taste Buds* Contain taste receptor cells; replaced about every 1–2 weeks.

2. Supertaster: Person with many taste buds who experiences tastes more intensely.

F. **Where in the brain is taste processed?** Temporal lobe (gustatory cortex), plus the thalamus relay.

XV. **Olfaction:** Sense of smell.

A. **What is the process of olfaction?** Odorant molecules enter the nose → bind receptors → signal to olfactory bulb → olfactory cortex/limbic system.

1. Odorants: Airborne chemical molecules that we smell.

2. Olfactory Bulb: Brain structure that receives signals from the nose's receptors.

3. Olfactory Cortex: Region (temporal lobe) where smell is processed.

4. Limbic System: Links smell to emotion and memory (amygdala, hippocampus).

B. **What is unique about the sense of smell (in a neural sense)?** Only sense that skips the thalamus; goes straight to the olfactory cortex and limbic system.

XVI. **How are taste, smell, and memory related?** Smell and taste combine to create flavor, and smell connects directly to the limbic system, so scents trigger strong memories and emotions.

XVII. **Where in the brain is smell processed?** Olfactory cortex (temporal lobe) and limbic system.

Learning Target 1.6.E: Explain how the structures and functions of the *touch* sensory system relate to behavior and mental processes.

XVIII. **How do we sense touch?** Skin has receptors for 4 basic sensations: pressure, warmth, cold, and pain; others combine from these.

A. **Pressure:** Mechanoreceptors detect touch and pressure.

B. **Warmth:** Warmth receptors detect rising skin temperature.

C. **Cold:** Cold receptors detect falling skin temperature.

D. **Pain:** Nociceptors detect harmful stimuli.

XIX. **Where in the brain is touch processed?** Somatosensory cortex in the parietal lobe.

A. **Why are some parts of our body more sensitive than others?** More cortex space is devoted to them (e.g., lips, fingers) and they have more receptors.

Learning Target 1.6.F: Explain how the structures and functions of the *pain* sensory system relate to behavior and mental processes.

I. **What is pain?** Unpleasant signal that warns of damage and protects the body.

D. **How is pain a bottom-up process?** Nociceptors detect damage and send signals up to the brain.

E. How is pain a top-down process? The brain's attention, expectations and emotions change how pain is felt.

F. How is pain psychological? Mood, attention, stress and expectations (e.g., placebo) change pain perception.

1. Embodied Cognition Physical body sensations influence thoughts and judgments (e.g., physical warmth → feeling socially warm).

A. How is pain social-cultural? Social context and culture shape how we express and experience pain (e.g., others' reactions).

B. How is pain influenced by biology? Pain involves nociceptors, spinal cord, brain circuits, and natural chemicals like endorphins.

1. Nociceptor: Sensory receptor (free nerve endings) that detects painful, harmful stimuli.

2. Pain Circuit: Nociceptors → spinal cord → thalamus → somatosensory cortex.

3. Gate-Control Theory: Spinal cord has a 'gate' that blocks or allows pain signals; non-pain input (rubbing) or brain signals can close it.

   (1) Phantom-Limb Pain - Pain felt in a missing limb because the brain still sends and interprets signals for it.

1. How do endorphins relate to pain? Natural painkillers that reduce pain and boost pleasure.

***C.* How do we control pain?** Distraction, relaxation/meditation, acupuncture, medication, exercise, hypnosis, and gate-closing techniques (massage).

Learning Target 1.6.G: Explain how the structures and functions that maintain balance (*vestibular*) and body movement (*kinesthetic*) relate to behavior and mental processes.

I. How do we sense our body's position and movement?

A. Proprioception: Sense of body position and movement in space.

B. Vestibular Sense: Sense of balance and head position/movement.

1. Semicircular Canals: Fluid-filled tubes in the inner ear that detect head rotation.

2. Vestibular Sacs: Detect head position and linear movement/gravity.

3. Vestibular Nerve: Carries balance info to the brain (cerebellum).

B. Kinesthetic Sense: Sense of body parts' position and movement via receptors in muscles, joints and tendons.

